

Image formation

Colour

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MIPT, spring 2024

Revise

Lecture 1: Biological vision

- Eyes structure and types
- Human eye parameters
- Binocular vision
- Ways to improve eyes performance
- Applications in science and business

Outline

1. Motivation
2. Color
3. Light geometry
4. Camera models
5. Sensors

Some basic definitions

- **Computer vision** (CV) is an interdisciplinary field that deals with how computers can be made to gain high-level understanding from digital images or videos
- **Image processing** and image analysis tend to focus on 2D images, how to transform one image to another, e.g., by pixel-wise operations such as contrast enhancement, local operations such as edge extraction or noise removal, or geometrical transformations such as rotating the image
- **Machine vision** (MV) is the technology and methods used to provide imaging-based automatic inspection and analysis for such applications as automatic inspection, process control, and robot guidance, usually in industry



Motivation

girafe
ai

01

Why do we need to understand imaging?

Because we can not get rid of physical part of imaging (both light formation and perception hardware)



Light and colour

girafe
ai

02

Mathematics of colour

In case of direct light

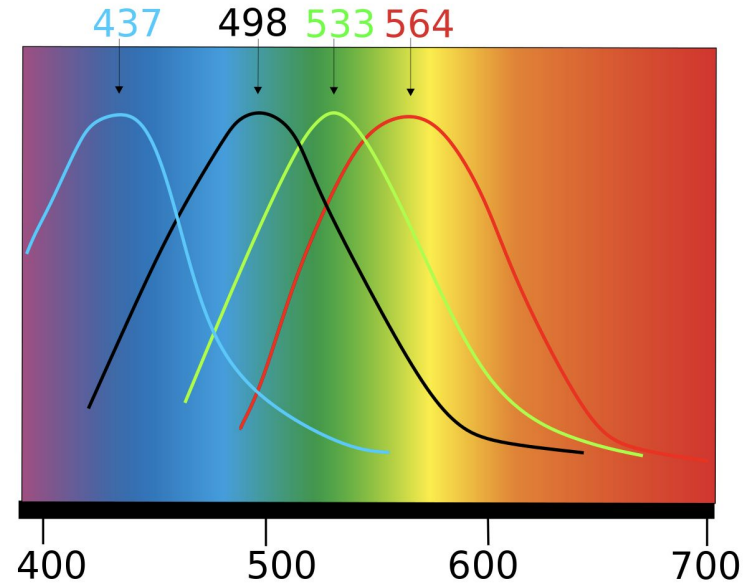
$$S(\lambda) \xrightarrow{\quad} \star \chi_i(\lambda)$$

short $\chi_B(\lambda)$

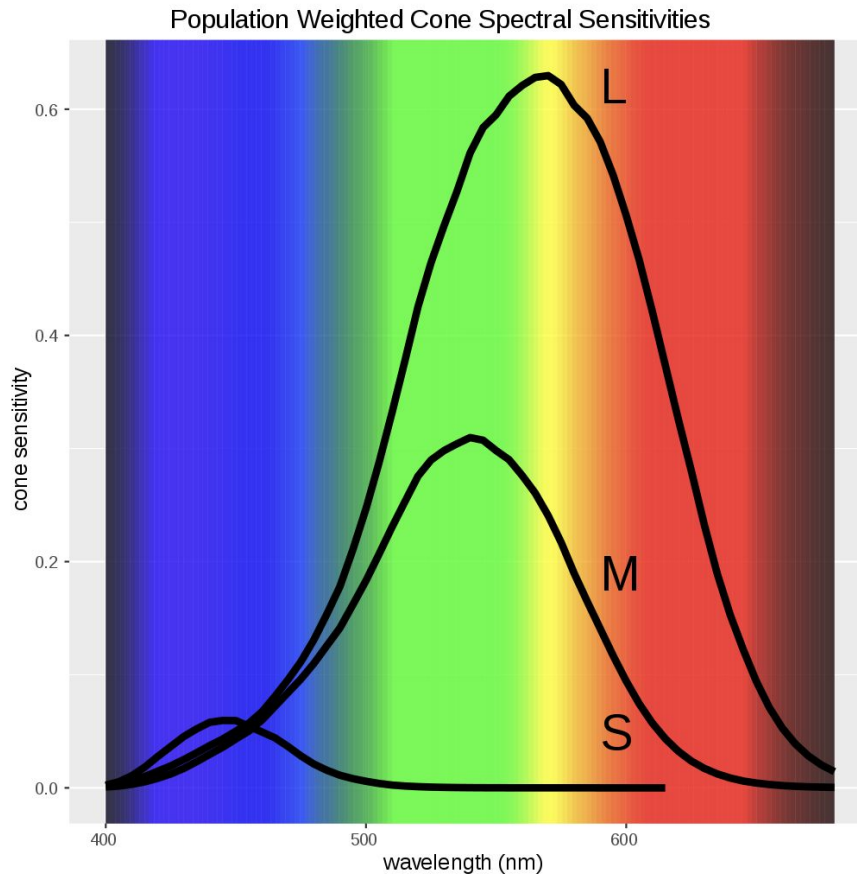
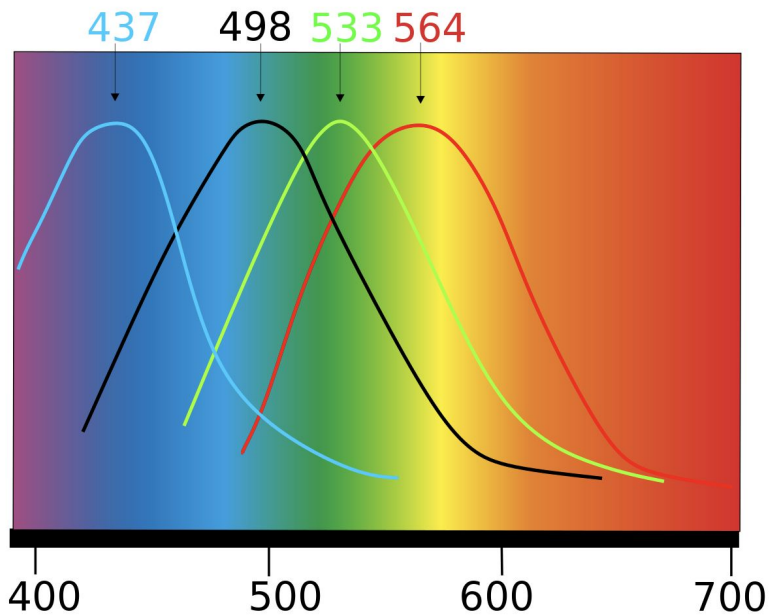
medium $\chi_G(\lambda)$

long $\chi_R(\lambda)$

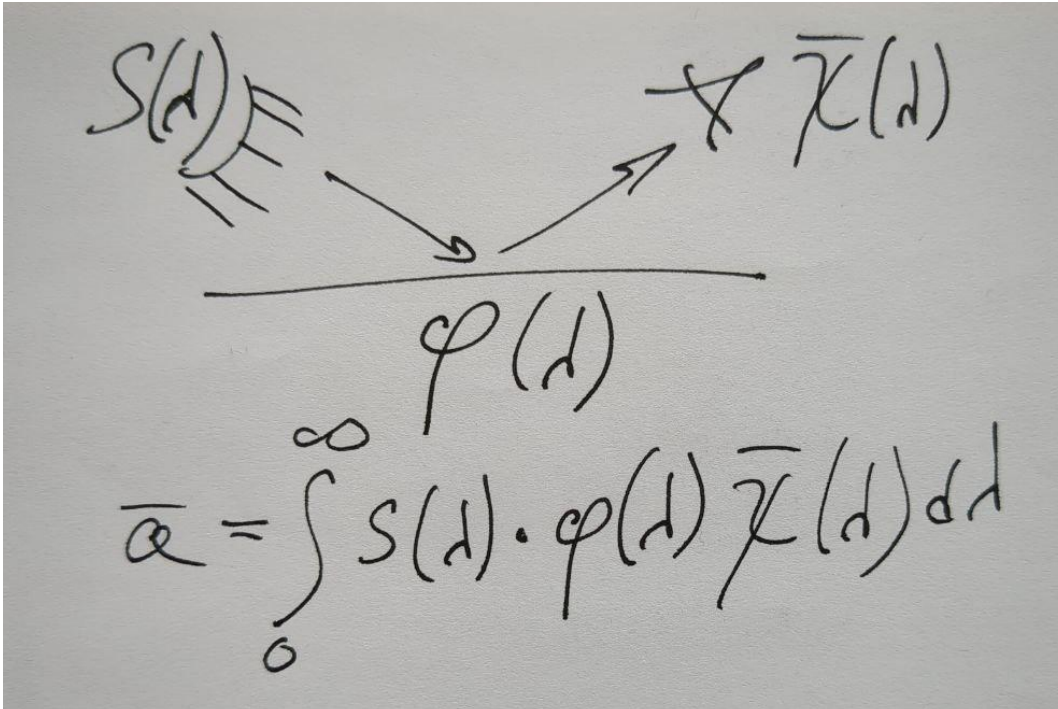
$$\bar{a} = \int_0^{\infty} S(\lambda) \cdot \bar{\chi}(\lambda) d\lambda$$



Sensitivity of cones



Reflected light



The diagram illustrates the reflection of light. An incident light beam, labeled $S(\lambda)$, is shown as a series of parallel lines with an arrow pointing towards a horizontal surface labeled $\varphi(\lambda)$. A reflected light beam, labeled $\bar{K}(\lambda)$, is shown as a series of parallel lines with an arrow pointing away from the same surface. Below the diagram, the mathematical expression for the average reflected light is given as:

$$\bar{Q} = \int_0^{\infty} S(\lambda) \cdot \varphi(\lambda) \bar{K}(\lambda) d\lambda$$

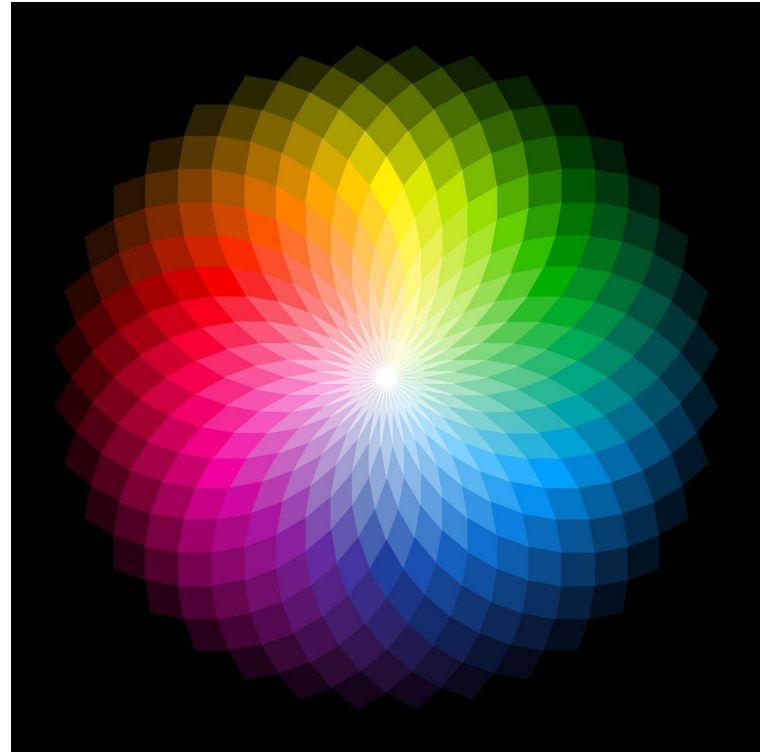


Colour

visual perception based on the electromagnetic spectrum.

One of pioneers of colour theory is Erwin Schrödinger

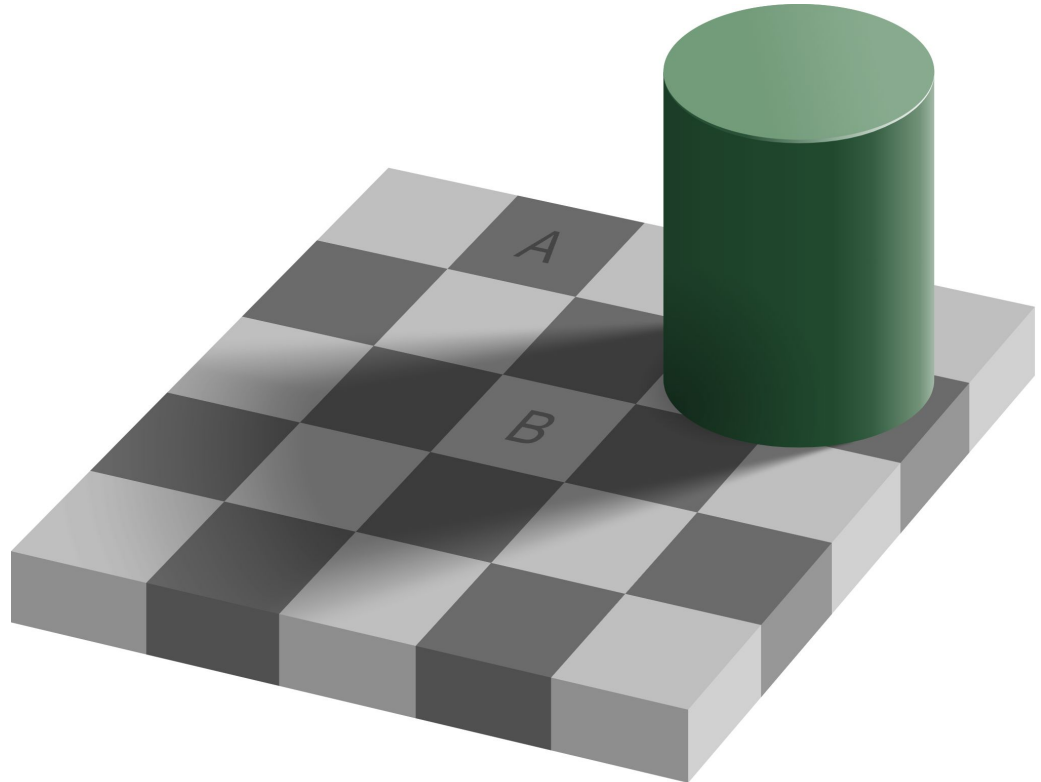
[Great lecture on colour theory \(Ru\)](#)



Color constancy

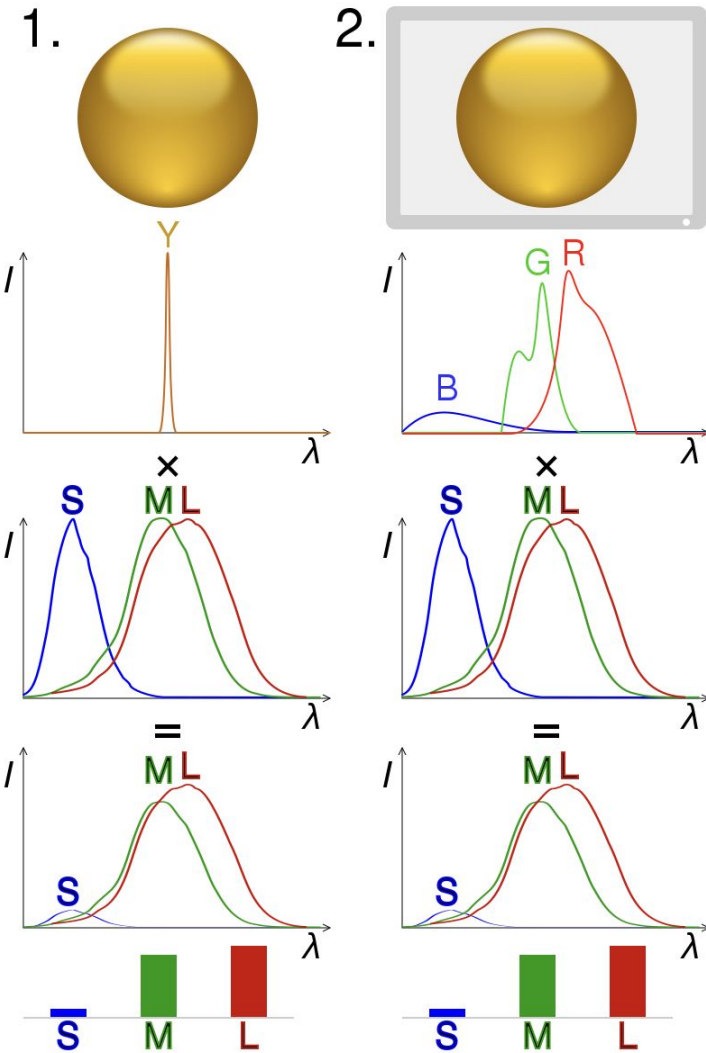
Despite wavelength we infer same colour of A and B

This is hard to achieve!

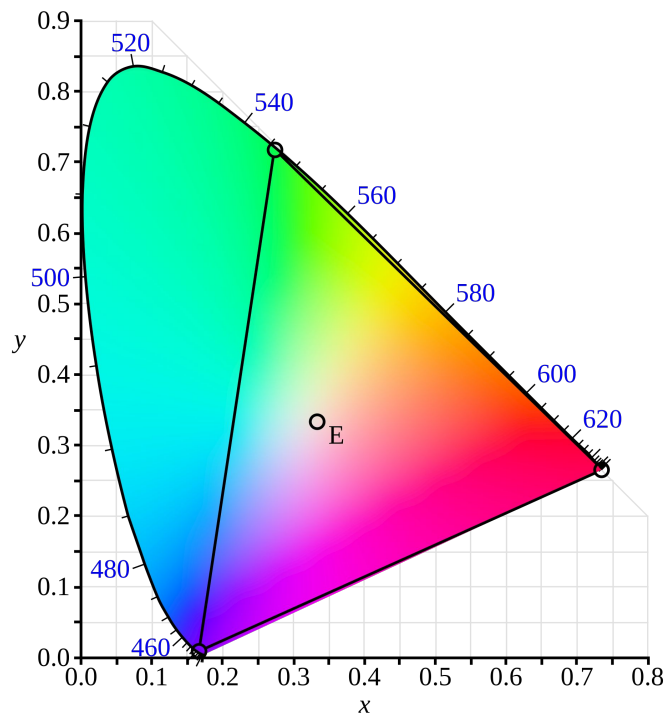


Metamerism

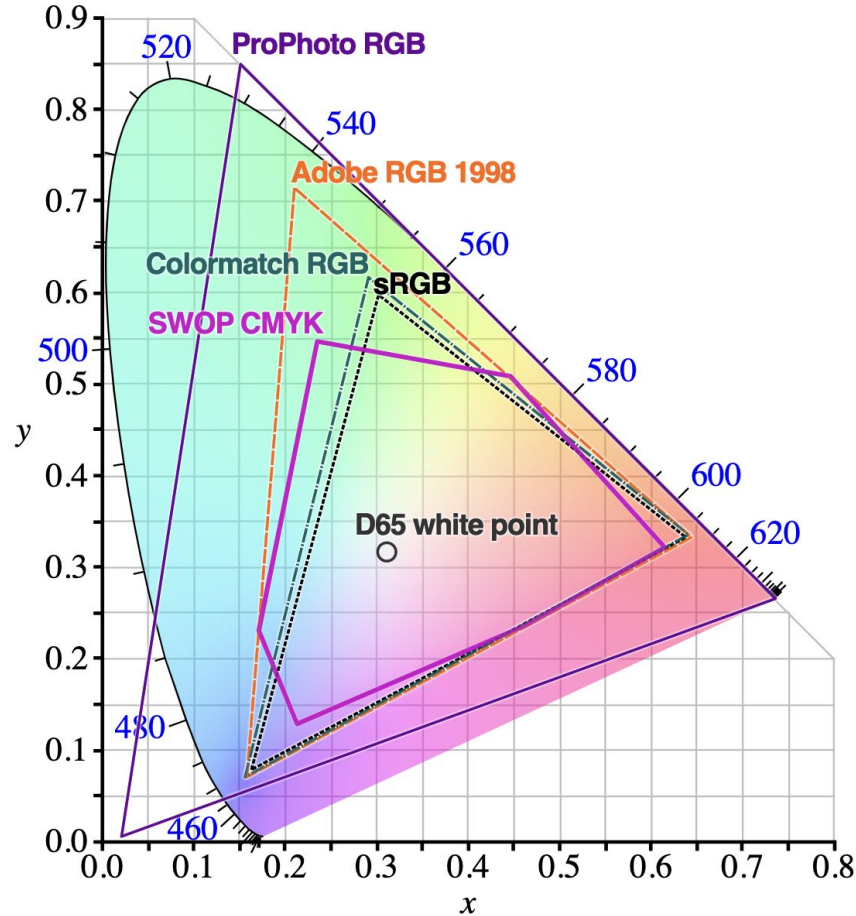
One feeling of colours can be produced by different spectrums



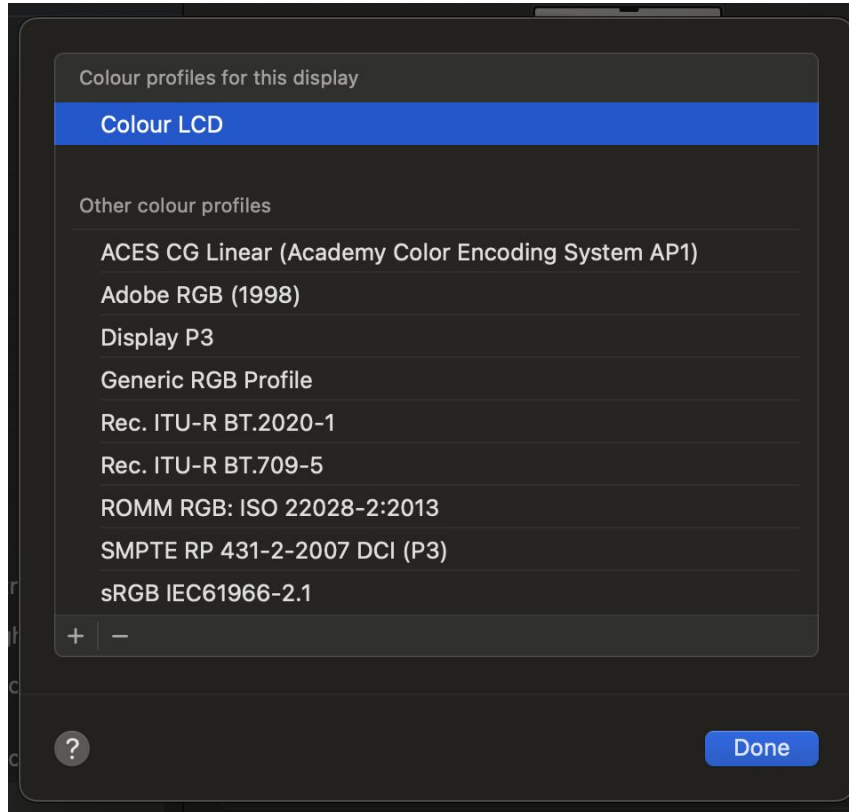
Colour space



CIE 1931

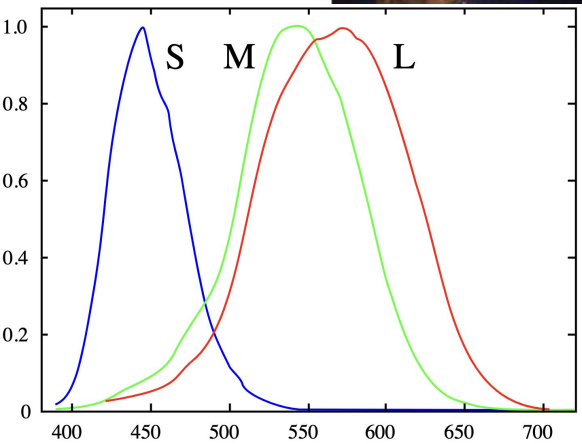
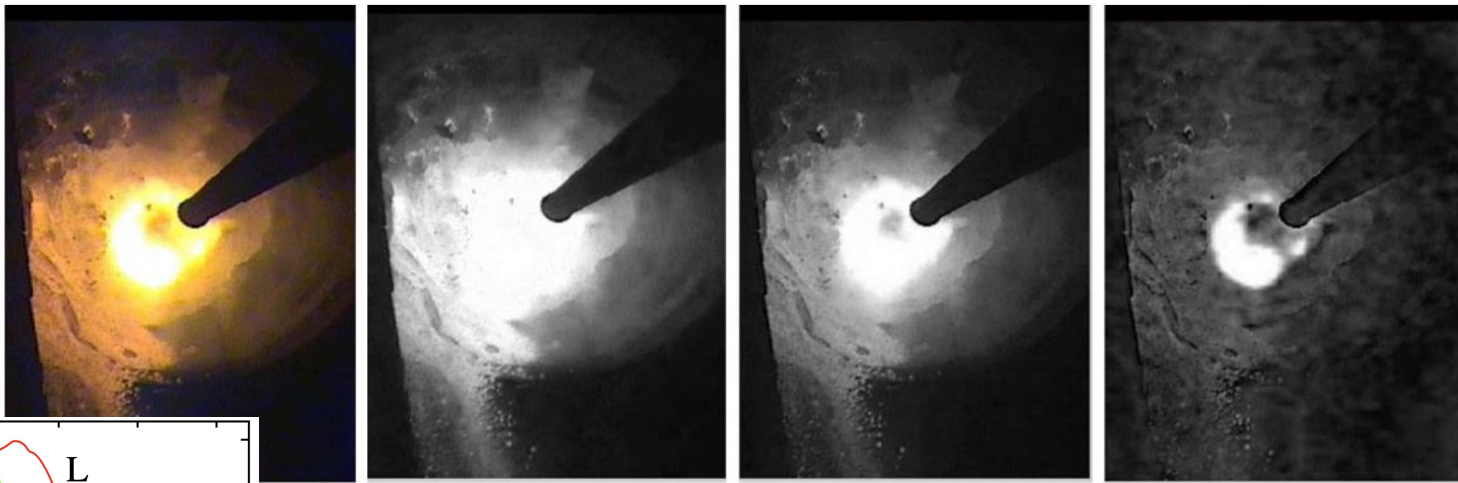


Colour space



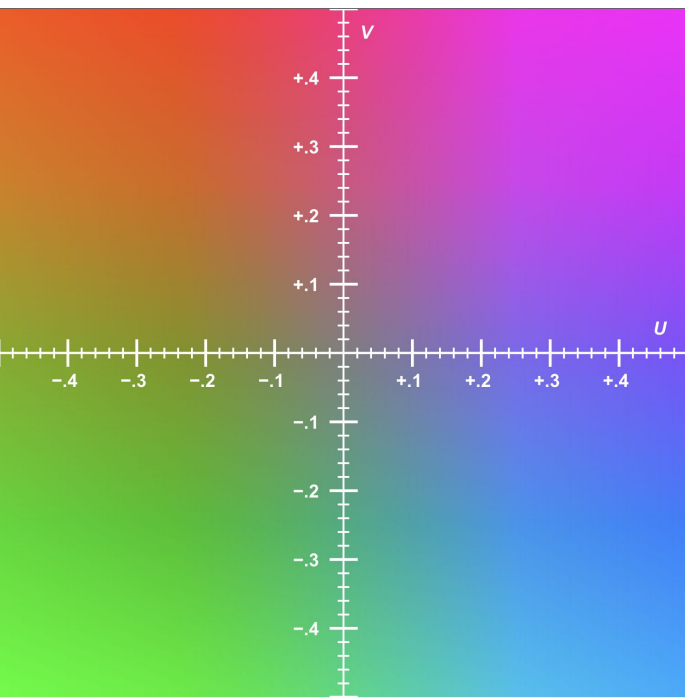
Colour spaces in
modern MacOS

Red Green Blue



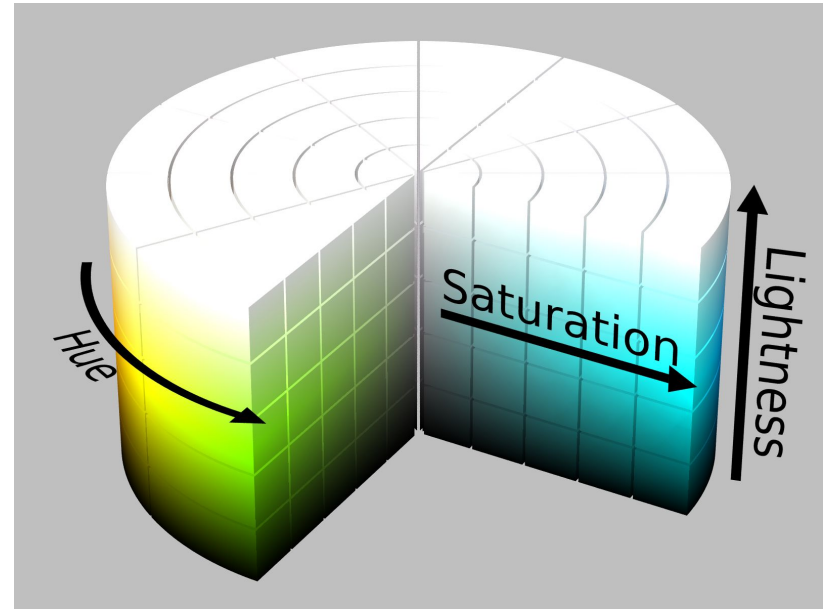
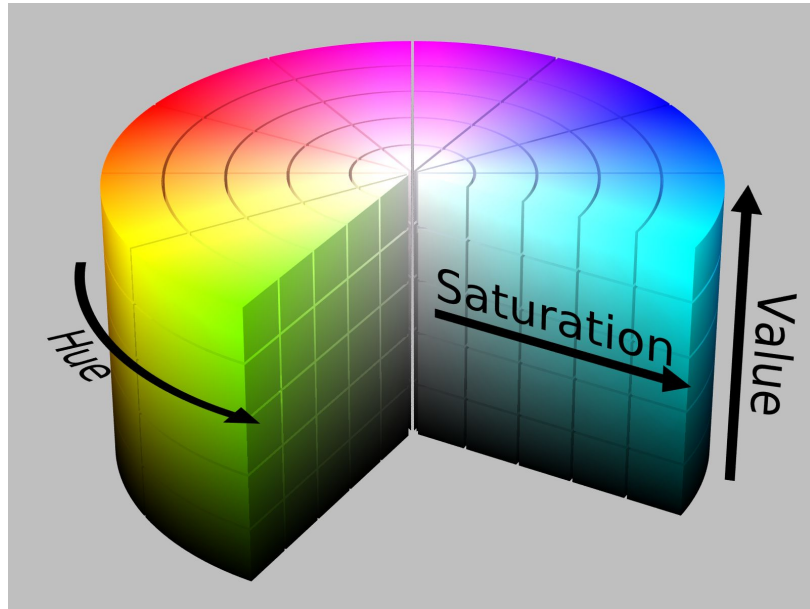
YUV

is used in television (analog signal) [wiki](#)



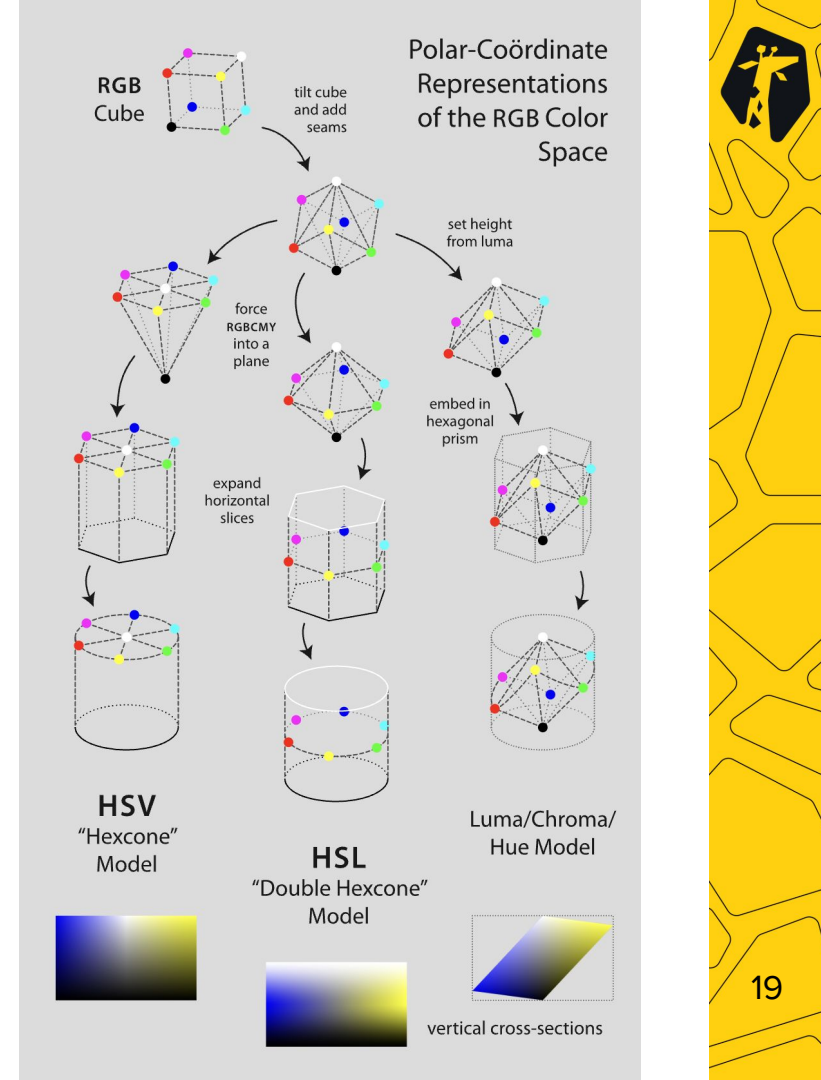
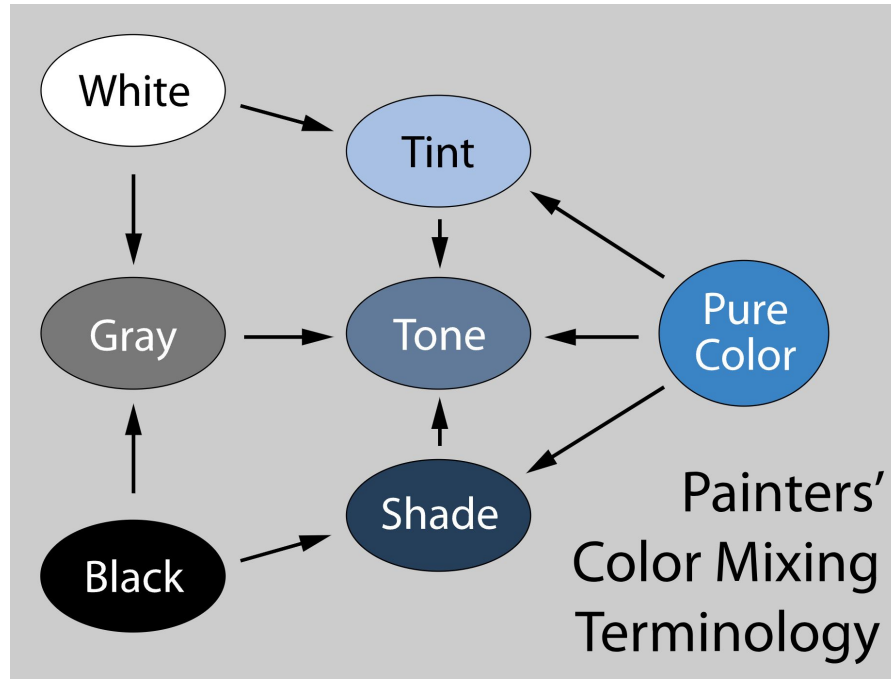
$$\begin{aligned}Y &= 0.2989 \cdot R + 0.5870 \cdot G + 0.1140 \cdot B, \\U &= 0.493 (B - Y), \\V &= 0.877 (R - Y).\end{aligned}$$

Hue, Saturation, Value (HSV)



HSV

animation



HSV usage



Fig. 13a. Color photograph (sRGB colorspace).



Fig. 13b. CIELAB L^* (further transformed back to sRGB for consistent display).



Fig. 13c. Rec. 601 luma Y' .



Fig. 13d. Component average: "intensity" I .



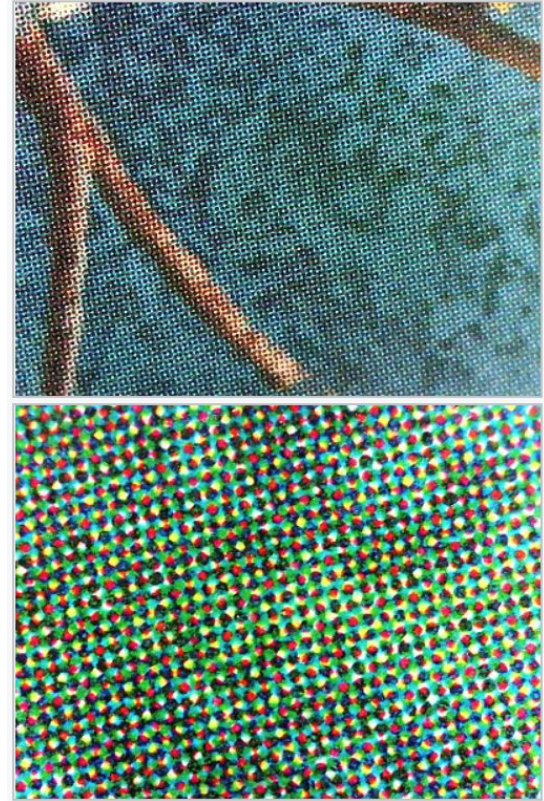
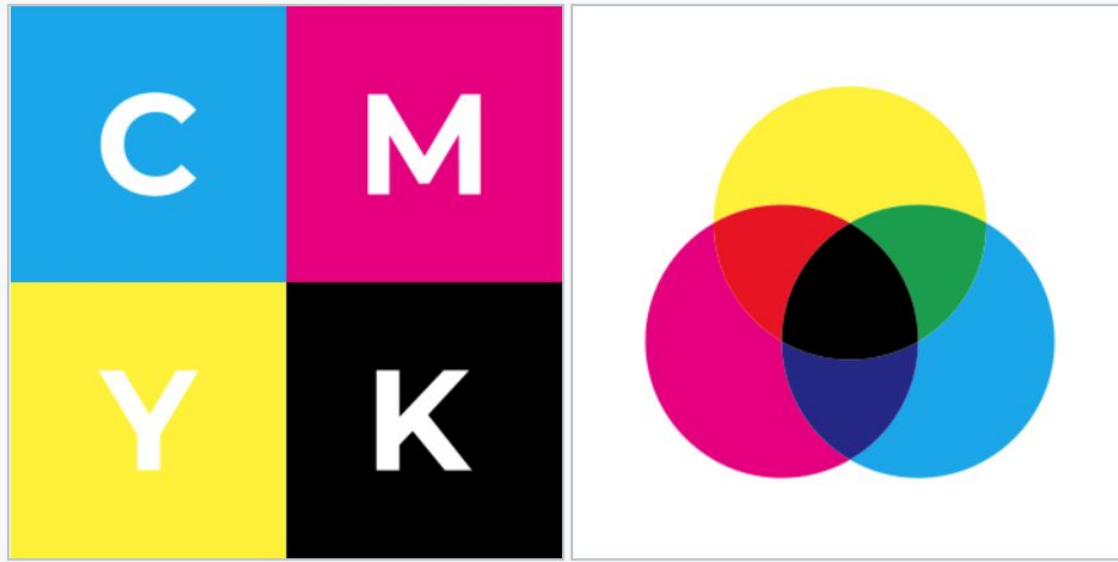
Fig. 13e. HSV value V .



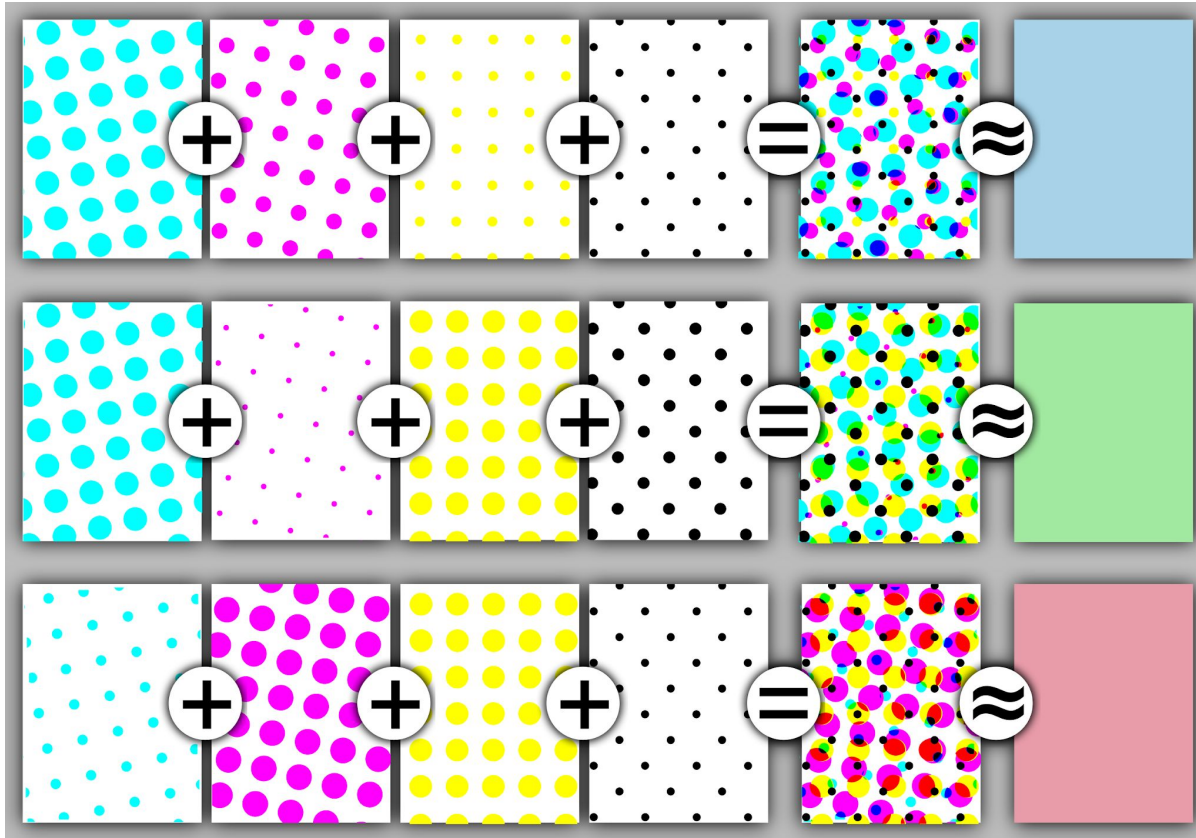
Fig. 13f. HSL lightness L .

CMYK

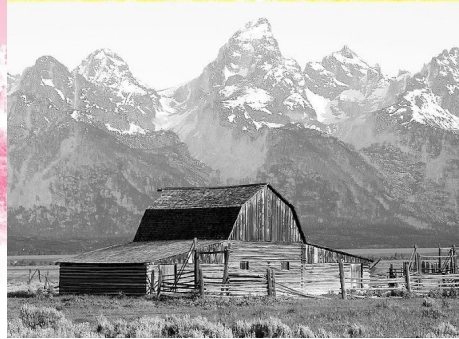
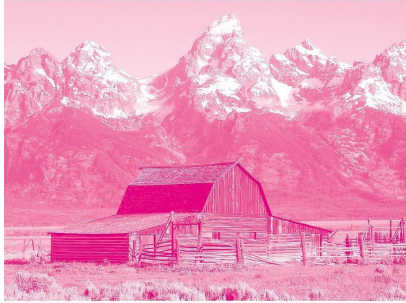
Cyan, Magenta, Yellow, Key (black)



CMYK printing



CMYK printing



Colour in Python

<https://github.com/colour-science/colour>

COLOUR SCIENCE

PROJECTS ▾

DOCUMENTATION ▾

BLOG ▾



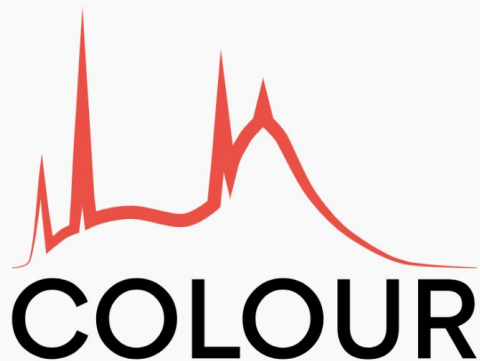
Colour is an open-source **Python** package providing a comprehensive number of algorithms and datasets for colour science.

It is freely available under the **BSD-3-Clause** terms.

Developed with  **JetBrains PyCharm**.

GET STARTED

📄 0.4.5 ALPHA

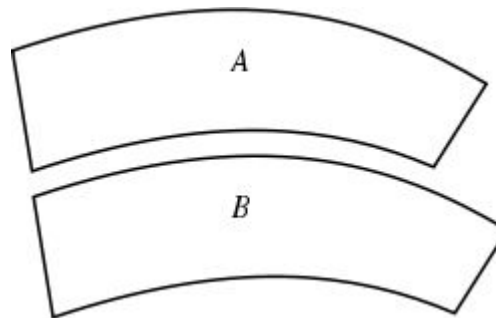
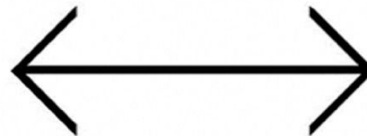
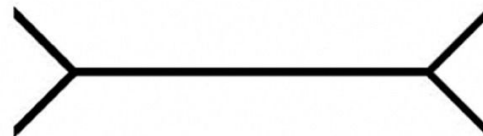
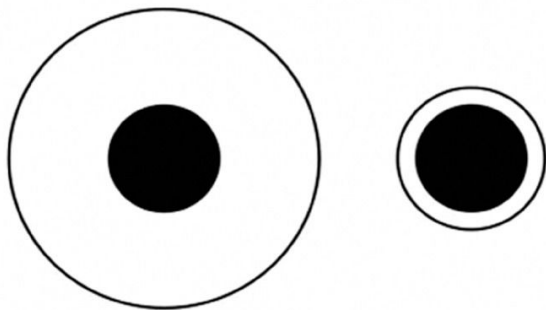
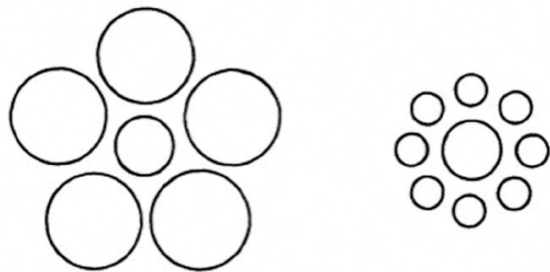


Light geometry

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ai

03

Size estimation



Perspective



(1) $\frac{AC \times BD}{BC \times AD} = \frac{A'C' \times B'D'}{B'C' \times A'D'}$

$\frac{(30 + 20) \times (20 + 10)}{20 \times (30 + 20 + 10)} = \frac{(7 + W)(W + 6)}{W(7 + W + 6)}$

$5W(W + 13) = 4(W + 7)(W + 6)$

$5W^2 + 65W = 4W^2 + 52W + 168$

$W^2 + 13W - 168 = 0$

$(W + 21)(W - 8) = 0$

$W > 0 \therefore W = 8 \text{ m}$

(2) $\frac{AC \times BV}{BC \times AV} = \frac{A'C'}{B'C'}$

$\frac{50 \times 90}{20 \times 120} = \frac{7 + W}{W}$

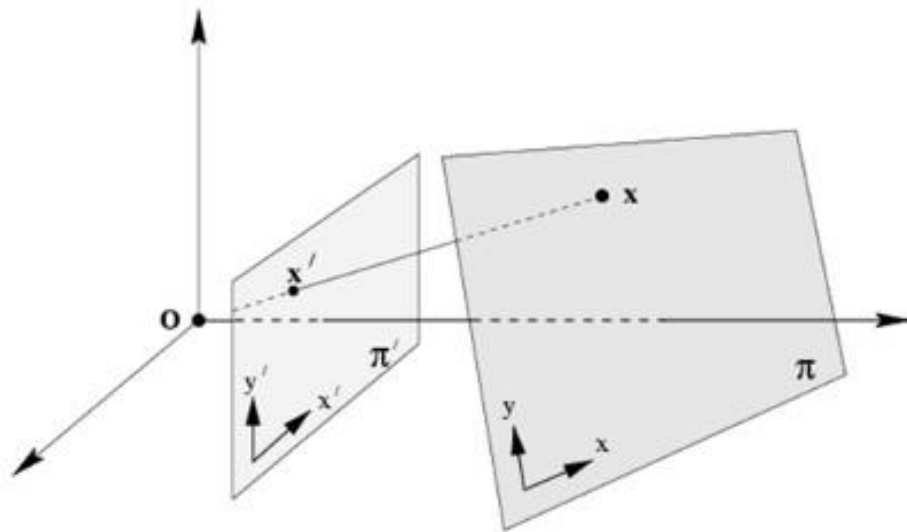
$15W = 8(7 + W)$

$7W = 56 \therefore W = 8 \text{ m}$

Homography

The homography matrix is a 3x3 matrix but with 8 DoF (degrees of freedom) as it is estimated up to a scale.

It is generally normalized with $h_{33}=1$



$$s \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \mathbf{H} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

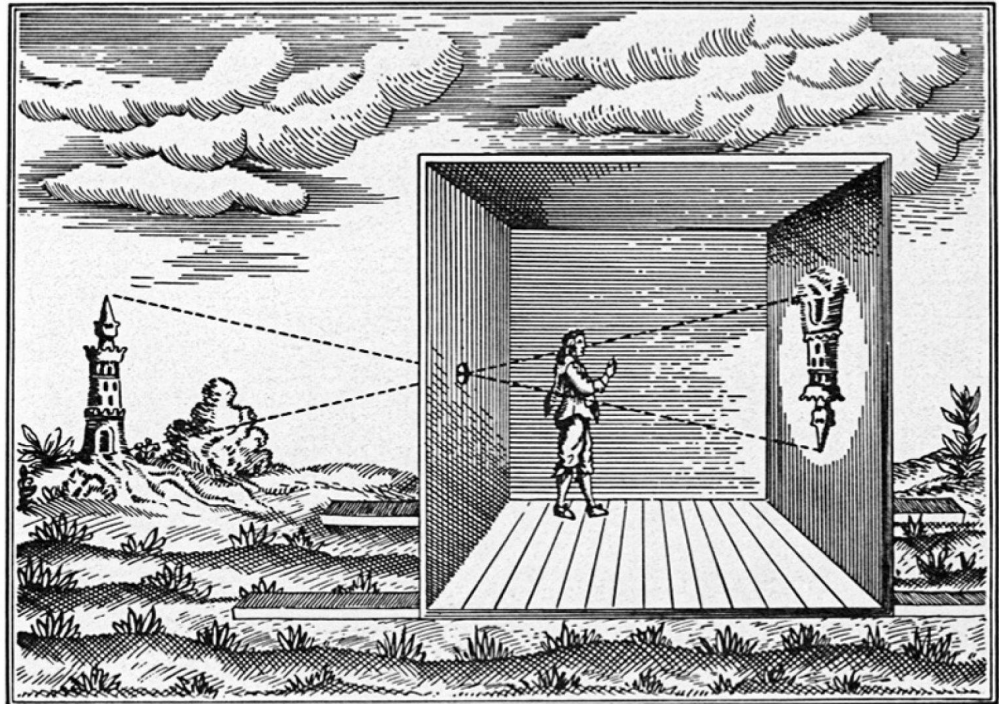
Camera models

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04

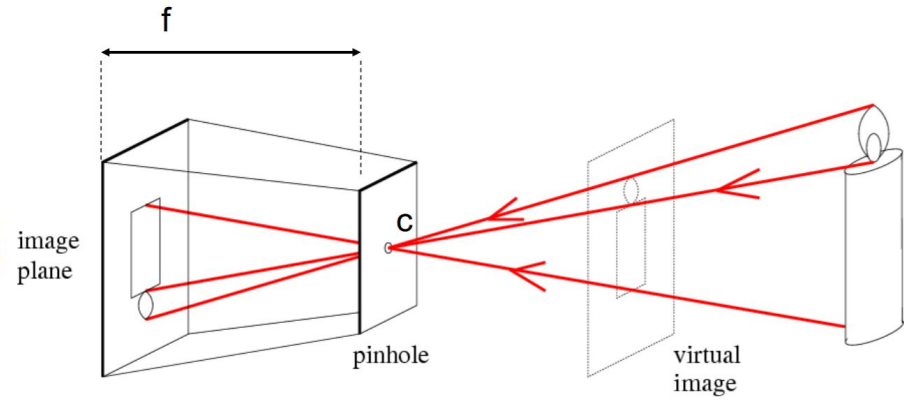
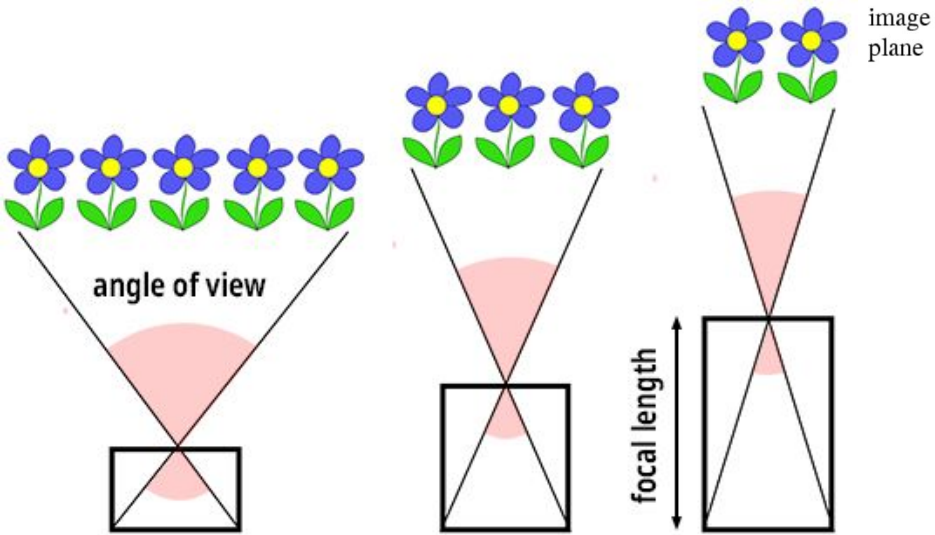
Camera obscura

- from Latin camera obscura - dark chamber
- used since 500 BC

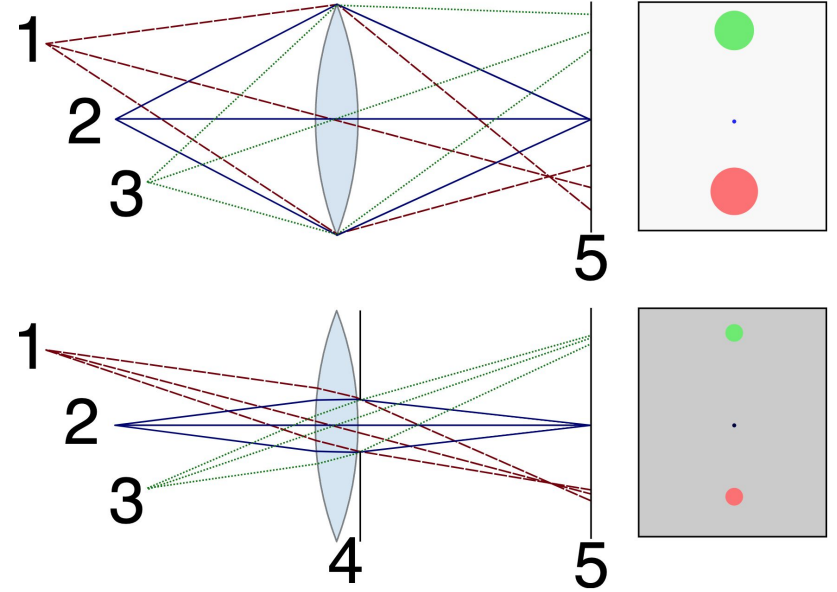
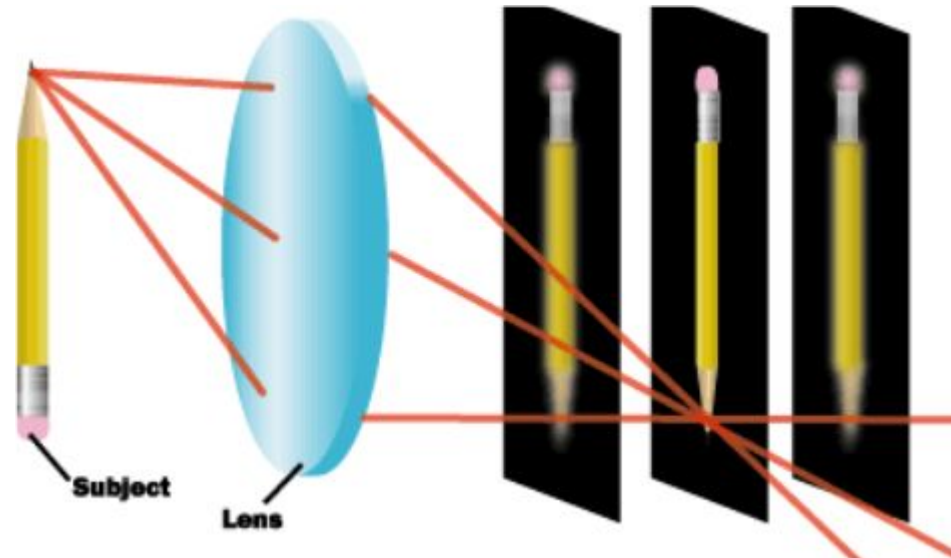


Pinhole camera model

Light don't refract



Single lens camera



Depth of field

distance between the nearest and the furthest objects that are in acceptably sharp focus in an image captured with a camera



Aperture = $f/1.4$. DOF = 0.8 cm



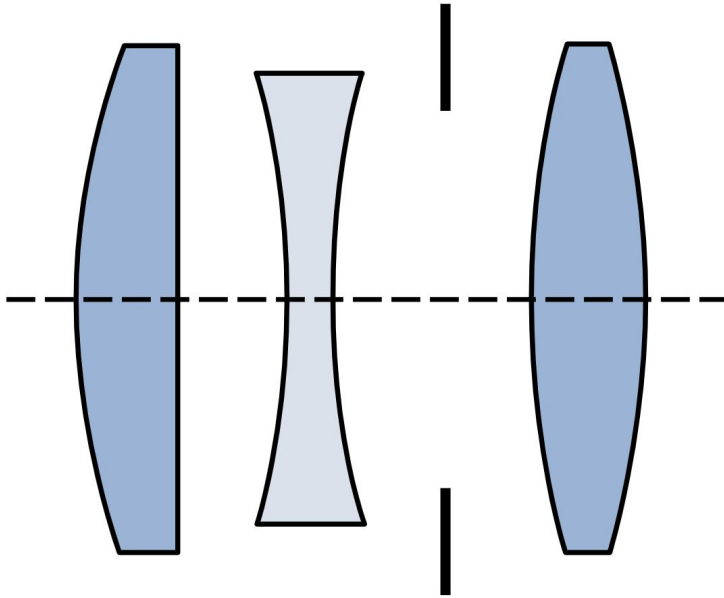
Aperture = $f/4.0$. DOF = 2.2 cm



Aperture = $f/22$. DOF = 12.4 cm



Compound lenses



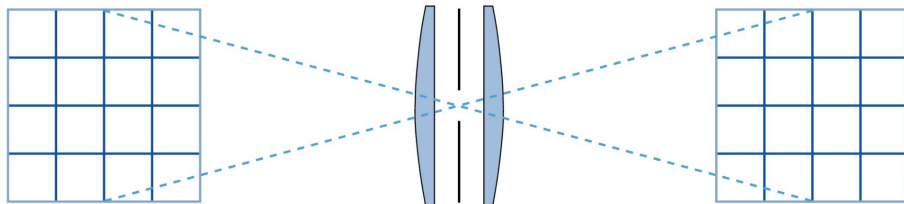
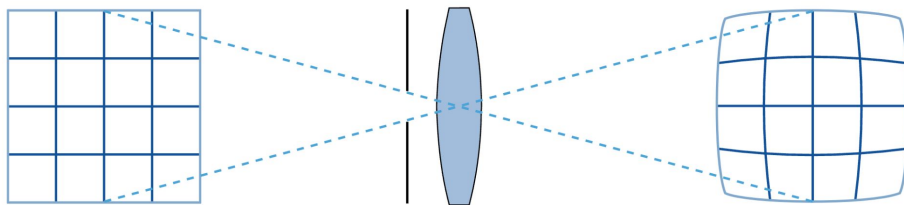
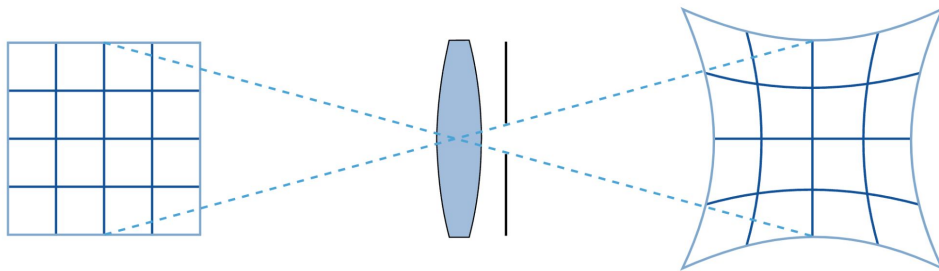
triplet lens



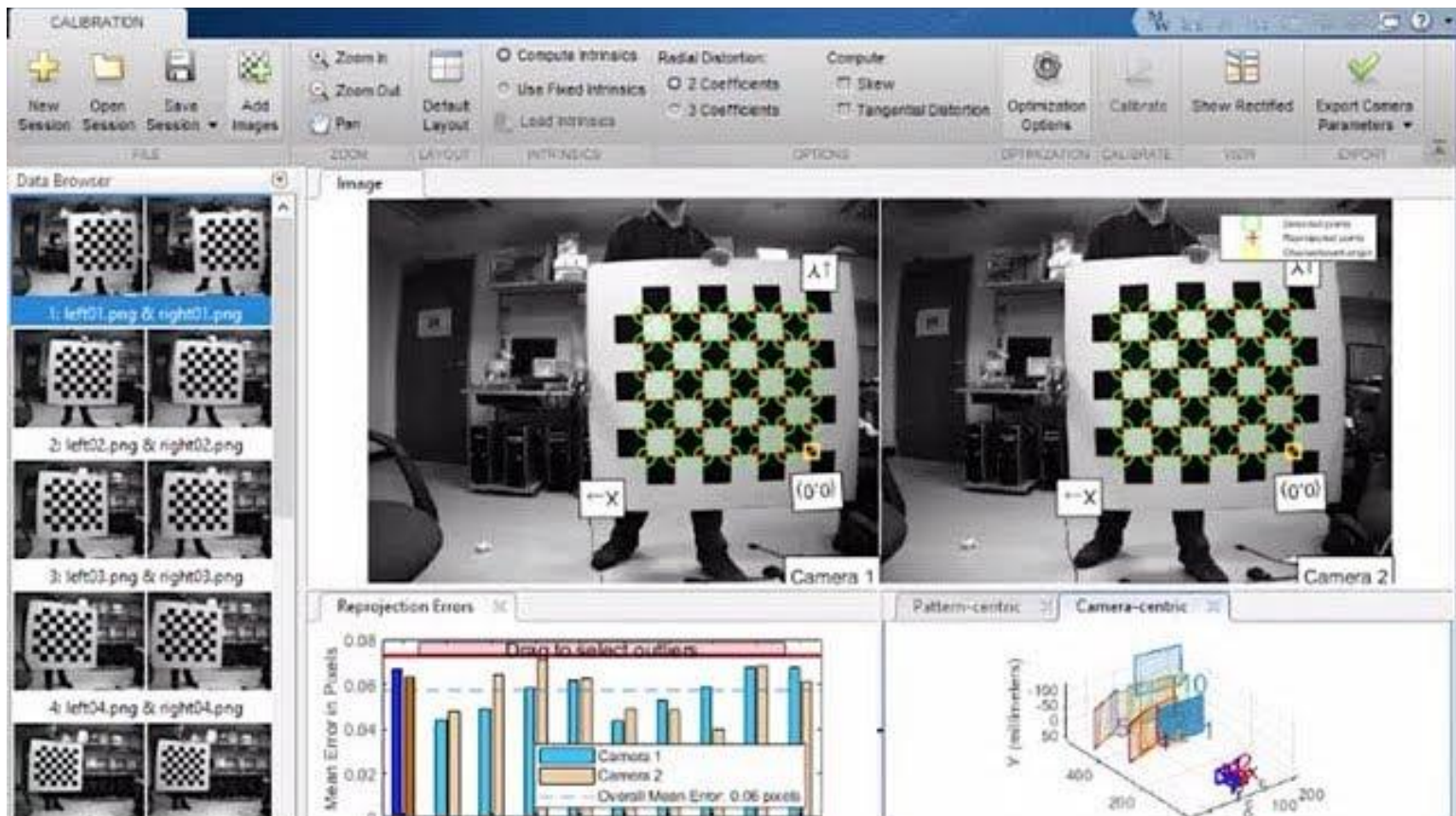
Distortion



Distortion

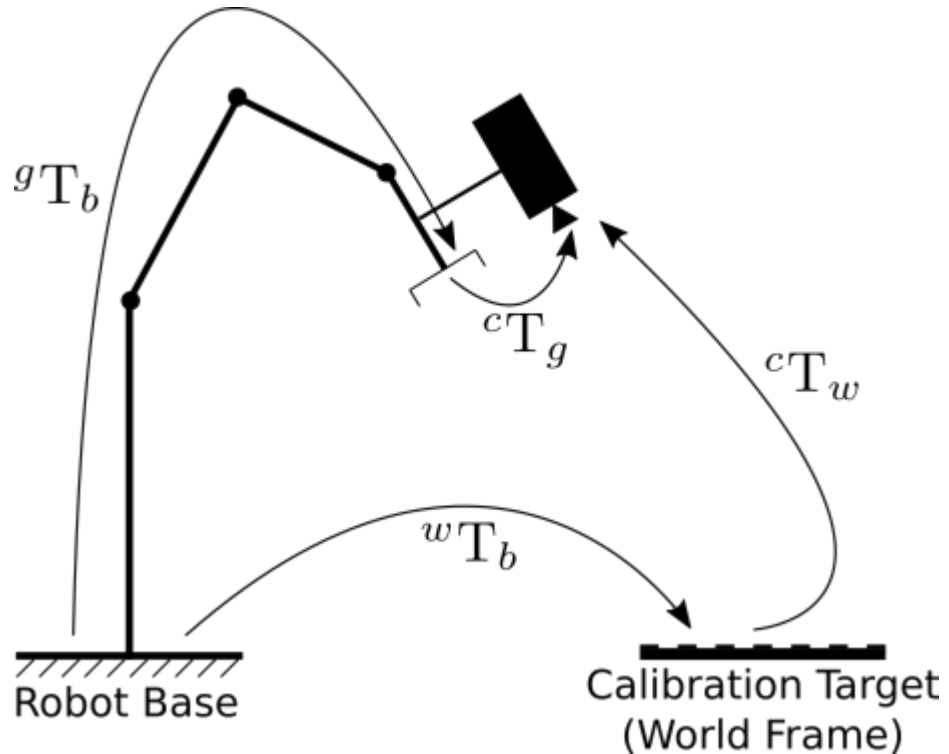


Camera resectioning



Camera model in OpenCV

[Documentation](#)



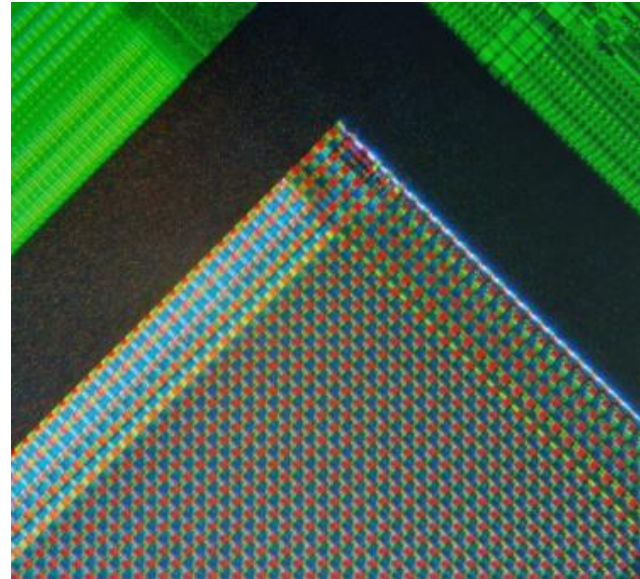
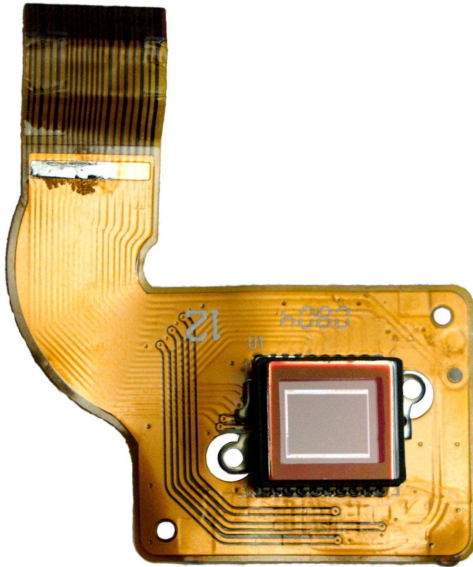
Sensors

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05

Image sensor

is a sensor that detects and conveys information used to form an image. It does so by converting the variable attenuation of light waves (as they pass through or reflect off objects) into signals, small bursts of current that convey the information



Charge-coupled device

- The charge-coupled device (CCD) was invented by Willard S. Boyle and George E. Smith at Bell Labs in **1969**
- in **2009** they were awarded the Nobel Prize for Physics[20] for their invention of the CCD concept

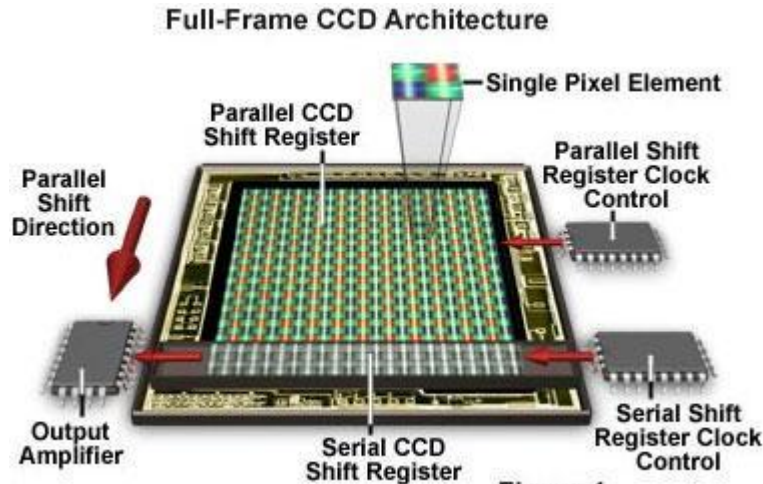
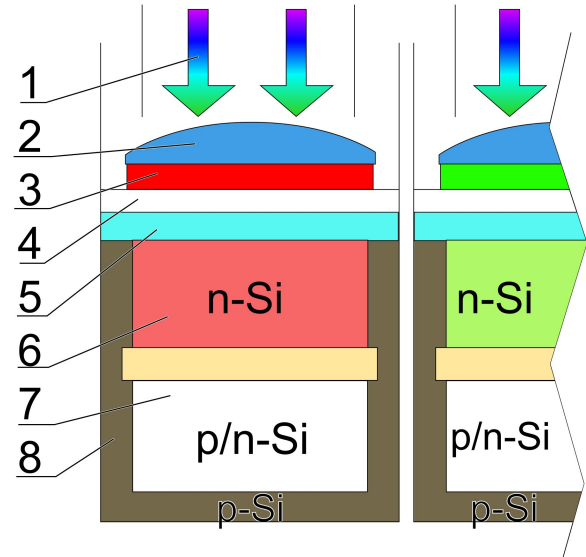
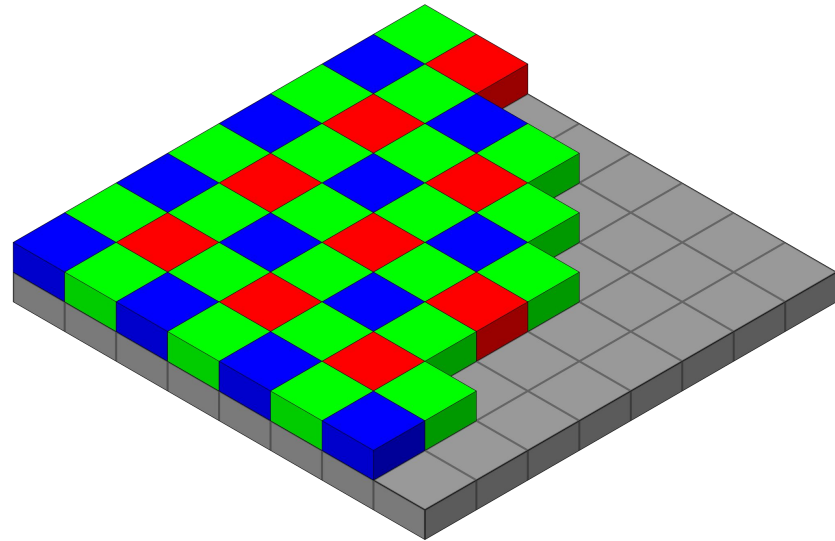
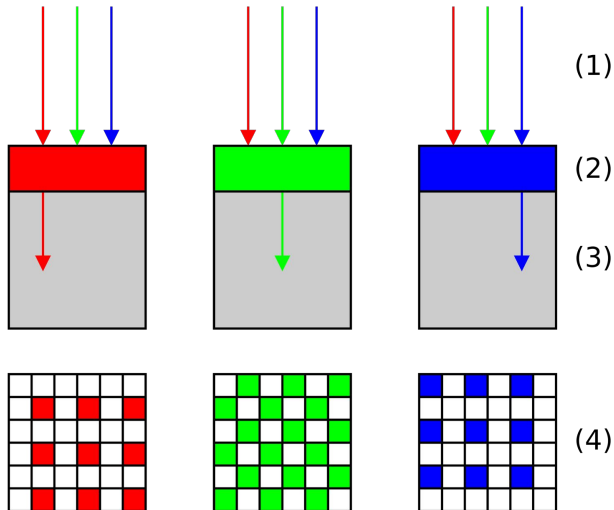


Figure 1



Bayer filter

is a **color filter array (CFA)** for arranging RGB color filters on a square grid of photosensors. Its particular arrangement of color filters is used in most single-chip digital image sensors used in digital cameras, camcorders, and scanners to create a color image. First patented in 1976



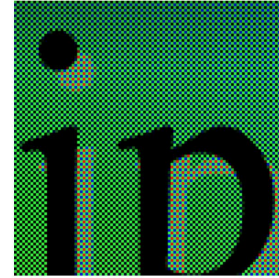
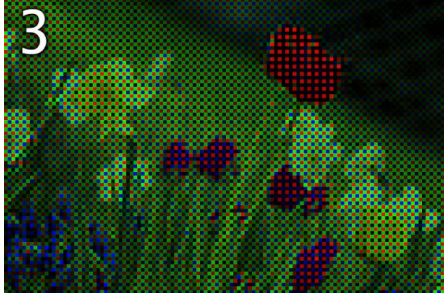
Different color filter arrays



	Bayer filter
	RGBE filter
	RYYB filter

	CYYM filter
	CYGM filter
	RGBW Bayer

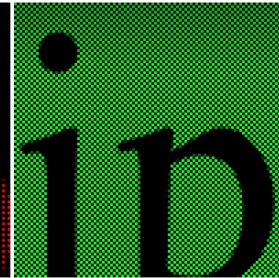
Demosaicing aka debayering



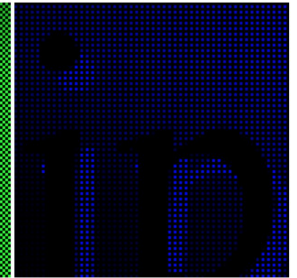
Bayer filter samples



Red



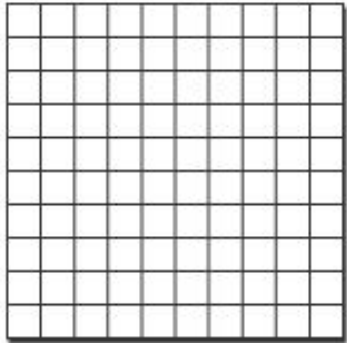
Green



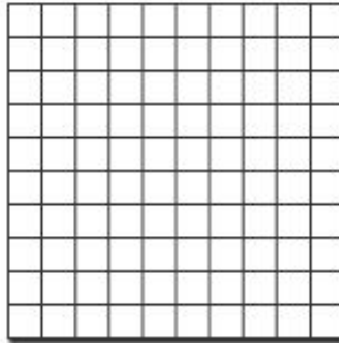
Blue

Camera shutter

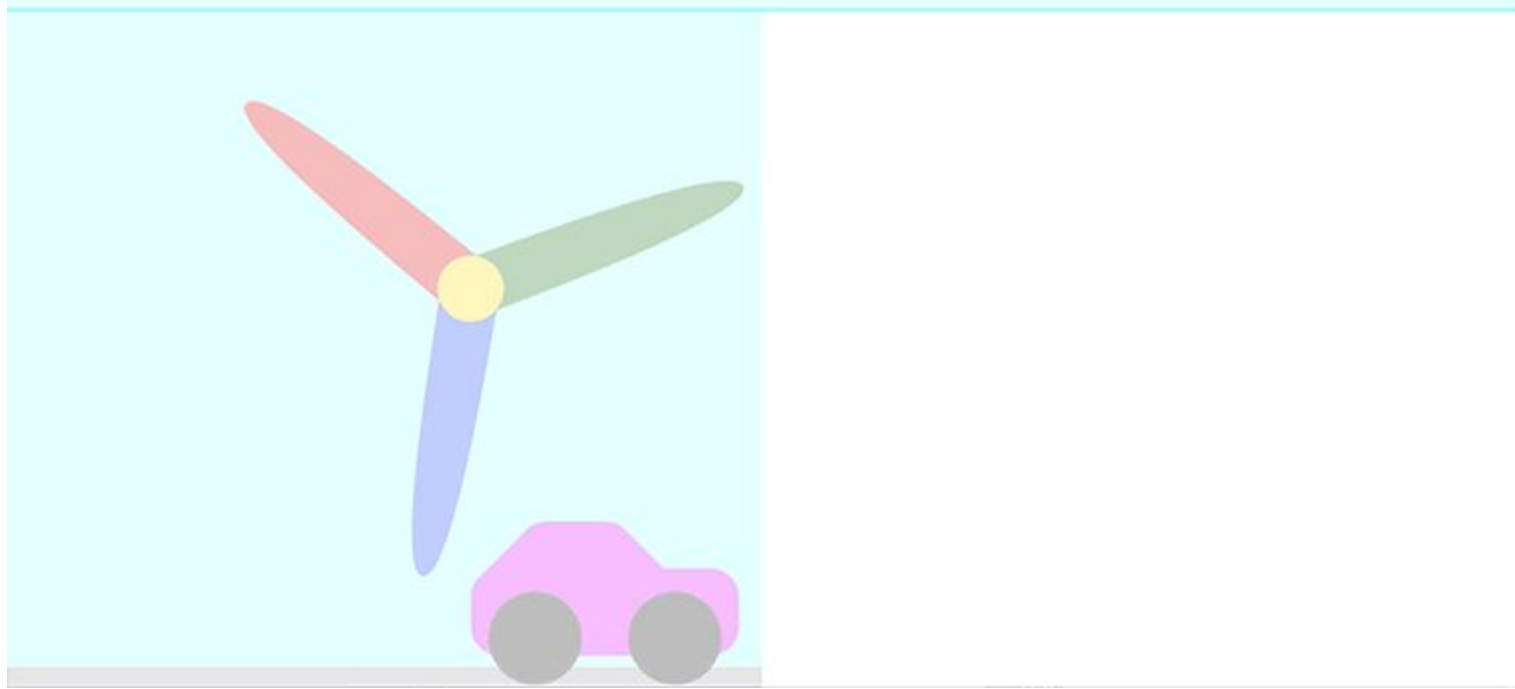
Rolling Shutter



Total Shutter



Rolling shutter



Comparison

[Global Shutter]



[Rolling Shutter]



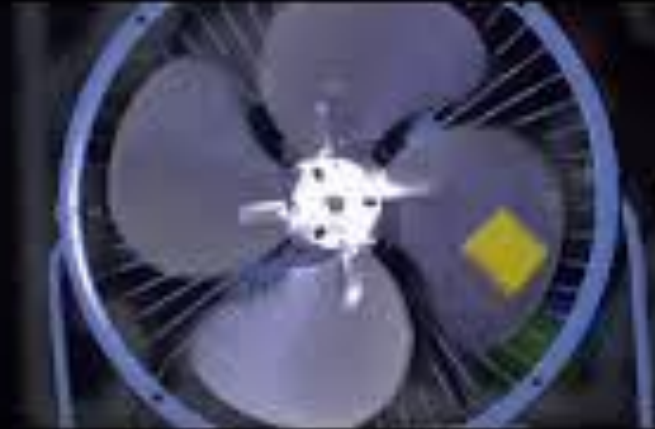
Frequency influence



Rolling shutter



Global shutter



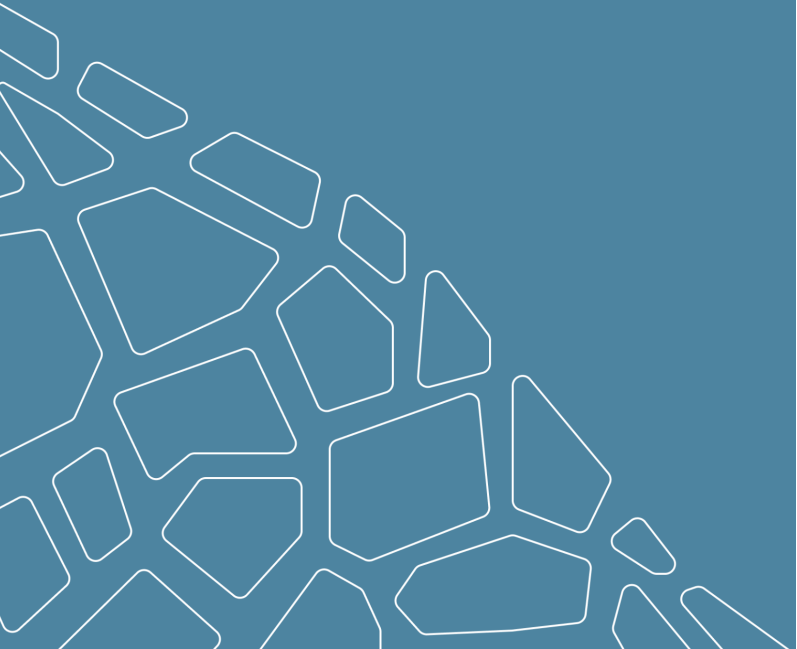
Shutter speed : 1/2000

Camera shutter

Сравнительная таблица плюсов и минусов

Режим	Rolling Shutter	Global Shutter
Снимок экспозиции	нет	да
Порядок считывания	Нет — очень разная «временная» последовательность воздействия	Да — чрезвычайно похожая последовательность воздействия
Временная задержка между различными регионами изображения области	Нет — разница до 10 мс (560 МГц) между центром и верхом или низом изображения	Да — все пиксели представляют одинаковое время экспозиции.
Возможность синхронизации	Комплекс для синхронизации. Требуется источник стробоскопического света. Более длительное время цикла.	Просто синхронизировать. Любой источник света. Более короткое время цикла.
Быстрая двойная экспозиция	нет	да
Максимальная частота кадров	Максимально доступно (не синхронизировано).	Максимальная частота кадров уменьшается вдвое.
Шум	Низкий уровень в районе 1 e^- до $1,3 \text{ e}^-$	Более высокий уровень в районе $2,3 \text{ e}^-$ до $2,6 \text{ e}^-$
Искажения, артефакты	Возможны	Нет
Эффективность рабочего цикла	Уменьшено, например, если требуется отключить подсветку во время «переходных» фаз считывания	Как правило, намного больше, так как не требуется переходной фазы считывания.

Revise

A decorative pattern of white-outlined, irregular polygons is located in the bottom-left corner of the slide, overlapping the blue background.

1. Motivation
2. Color
3. Light geometry
4. Camera models
5. Sensors

Thanks for attention!

Questions?

